

Spring 1982

Problem 1 (Sp77, Sp82, Sp20). Prove the Fundamental Theorem of Algebra: Every nonconstant polynomial with complex coefficients has a complex zero.

Problem 2 (Sp82). Let $S \subset \mathbb{R}^n$ be a subset which is uncountable. Prove that there is a sequence of distinct points in S converging to a point of S .

Problem 3 (Sp82). Let A and B be $n \times n$ complex matrices. Prove that

$$|\operatorname{tr}(AB^*)|^2 \leq \operatorname{tr}(AA^*) \operatorname{tr}(BB^*)$$

Problem 4 (Sp82). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ have directional derivatives in all directions at the origin. Is f differentiable at the origin? Prove or give a counterexample.

Problem 5 (Sp82, Sp93). Let $\{g_n\}$ be a sequence of twice differentiable functions on $[0, 1]$ such that $g_n(0) = g'_n(0) = 0$ for all n . Suppose also that $|g''_n(x)| \leq 1$ for all n and all $x \in [0, 1]$. Prove that there is a subsequence of $\{g_n\}$ which converges uniformly on $[0, 1]$.

Problem 6 (Sp82). Suppose that $f(x)$ is a polynomial with real coefficients and a is a real number with $f(a) \neq 0$. Show that there exists a real polynomial $g(x)$ such that if we define p by $p(x) = f(x)g(x)$, we have $p(a) = 1$, $p'(a) = 0$, and $p''(a) = 0$.

Problem 7 (Sp82, Sp93). Suppose that the group G is generated by elements x and y that satisfy $x^5y^3 = x^8y^5 = 1$. Is G the trivial group?

Problem 8 (Sp77, Fa81, Sp82, Sp26). Evaluate $\int_{-\infty}^{\infty} \frac{\cos nx}{x^4 + 1} dx$.

Problem 9 (Sp78, Sp82, Su82, Fa90). Determine the Jordan Canonical Form of the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 4 \end{pmatrix}$$

Problem 10 (Sp82). Prove that any group of order 77 is cyclic.

Problem 11 (Sp82). Decide whether a finite limit

$$\lim_{z \rightarrow 0} \left(\frac{1}{\tan^2 z} - \frac{1}{z^2} \right)$$

exists, where z is a complex variable, and if yes, compute the limit.

Problem 12 (Sp82). Prove or give a counterexample: Every connected, locally pathwise connected set in $\mathbb{R}^n$ is pathwise connected.

Note: See also ??.

Problem 13 (Sp82). Let $T : V \rightarrow W$ be a linear transformation between finite dimensional vector spaces. Prove that

$$\dim(\ker T) + \dim(\operatorname{img} T) = \dim V$$

Problem 14 (Sp82). Let $f : I \rightarrow \mathbb{R}$, where I is an interval of $\mathbb{R}$, such that $f(x) > 0$, $x \in I$. Suppose that $e^{cx} f(x)$ is convex in I for every real number c . Show that $\log f(x)$ is convex in I .

A function $g : I \rightarrow \mathbb{R}$ is *convex* if

$$g(tx + (1-t)y) \leq tg(x) + (1-t)g(y)$$

for all x and y in I and $0 \leq t \leq 1$.

Problem 15 (Su79, Sp82, Sp83, Su84, Fa91, Fa98, Sp11, Fa15, Fa16, Fa23). Let $\mathbb{F}$ be a finite field with q elements and let V be an n -dimensional vector space over $\mathbb{F}$.

1. Determine the number of elements in V .
2. Let $\operatorname{GL}_n(\mathbb{F})$ denote the group of all $n \times n$ nonsingular matrices over $\mathbb{F}$. Determine the order of $\operatorname{GL}_n(\mathbb{F})$.
3. Let $\operatorname{SL}_n(\mathbb{F})$ denote the subgroup of $\operatorname{GL}_n(\mathbb{F})$ consisting of matrices with determinant 1. Find the order of $\operatorname{SL}_n(\mathbb{F})$.

Problem 16 (Sp82, Sp93). Prove that if G is a group containing no subgroup of index 2, then any subgroup of index 3 is normal.

Problem 17 (Sp82, Fa08, Sp11, Sp25). For each of the following statements, either prove it or give a counterexample:

1. If $f_n(x)$ are continuous real valued functions on the unit interval, and f_n converges pointwise to the function f , that is, $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ for all x , then

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) \, dx = \int_0^1 f(x) \, dx$$

2. If f_n are continuous and converges uniformly to a function f then the sequence $\int_0^1 f_n(x) dx$ has limit $\int_0^1 f(x) dx$.
3. If $g(m, n)$ is real for all integers m, n , and

$$\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} g(m, n) \quad \text{and} \quad \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} g(m, n)$$

are both defined, then they are equal.

4. If the real valued functions $h_n(x)$ are continuous on the unit interval, and $\lim_{n \rightarrow \infty} h_n(x) = h(x)$ for all x , then $h(x)$ is a continuous function of x .

Problem 18 (Su79, Sp82, Sp91, Sp96). Let f be a continuous complex valued function on $[0, 1]$, and define the function g by

$$g(z) = \int_0^1 f(t)e^{tz} dt \quad z \in \mathbb{C}$$

Prove that g is analytic in the entire complex plane.

Problem 19 (Su77, Su80, Sp82, Sp83). Prove that the initial value problem

$$\frac{dx}{dt} = 3x + 85 \cos x \quad x(0) = 77$$

has a solution $x(t)$ defined for all $t \in \mathbb{R}$.

Problem 20 (Sp82, Sp11, Fa17, Sp21). Prove that the polynomial $x^4 + x + 1$ is irreducible over $\mathbb{Q}$.